Assistant Professor Exam – 2024

ECONOMICS

UNIT – 3 : GROWTH & DEVELOPMENT

Detailed Notes with Diagrams (Hinglish)

Covers every topic of the official syllabus:

Concept, Characteristics & Indicators of Economic Growth; Inclusive & Sustainable Development

Circular Causation; Structural View of Underdevelopment; Measurement of Development (HDI, Quality of Life)

Environmental Sustainability; Environment & Development Dilemma; Environmental Accounting & Regulation in India

Climate Change – Impact & Remedies; Coase Bargaining Solutions

Growth Models: Robinson, Kaldor, Hicks & Harrod, Meade, Lewis, Ranis-Fei; Dependency Theory

Approaches: Balanced & Unbalanced Growth, Critical Minimum Effort, Big-Push, Low-Income Equilibrium Trap

Endogenous Growth – Education, Health, R&D, Knowledge; Cross-Country Differentials

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1. Concept, Characteristics & Indicators of Growth

1.1 Economic Growth vs Economic Development

Economic Growth ka matlab hai ek economy ke **real national income / real per capita income** mein ek lambe samay (long period) tak sustained increase. Ye ek **quantitative (sankhyatmak)** concept hai — sirf "kitna zyada produce hua" dekhta hai.

Economic Development zyada broad concept hai — ismein growth ke saath-saath **structural changes, distribution of income, poverty/unemployment reduction, health, education aur quality of life** mein improvement bhi shaamil hai. Ye **quantitative + qualitative (gunatmak)** dono hai. Yani:

> **Development = Growth + Structural/Institutional Change + Welfare improvement**

"Growth bina development possible hai (income badhe par garib aur garib ho), par development bina growth mushkil hai."

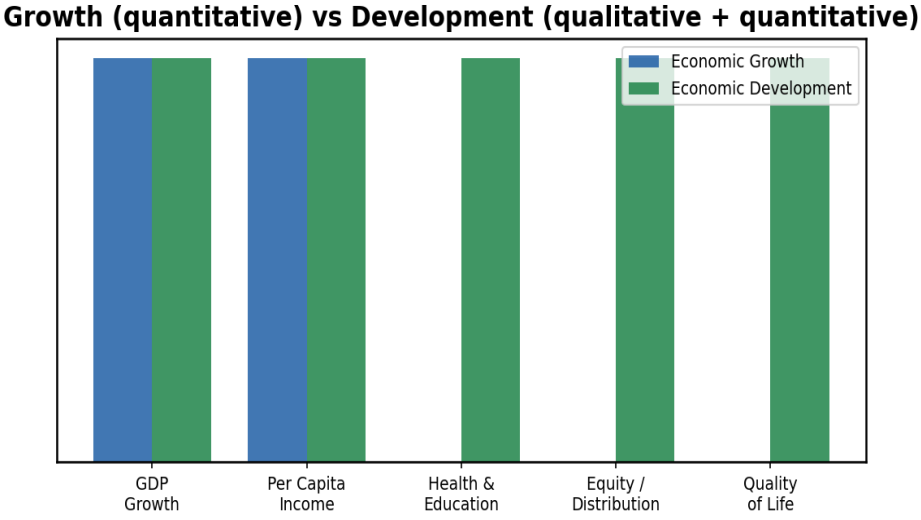


Fig 1.1 — Growth sirf output/income dekhta hai; development equity, health, education aur quality of life bhi.

Basis	Economic Growth	Economic Development
Nature	Quantitative	Quantitative + Qualitative
Scope	Narrow (income/output)	Broad (income + welfare + structure)
Measure	GDP, GNP, Per capita income	HDI, PQLI, poverty, inequality, etc.
Applies to	Mainly developed economies	Mainly developing economies
Change	More of same output	Structural transformation

1.2 Characteristics of Economic Growth

- **Sustained rise** in real per capita income over a long period (short-term ups ignore).
- **Structural transformation** — agriculture se industry aur services ki taraf shift (sectoral change).
- **Capital accumulation & technological progress** key drivers hain.
- **Rising productivity** of labour and capital.
- **Demographic changes** — urbanisation, falling birth/death rates.

1.3 Indicators of Growth & Development

Development measure karne ke liye sirf income kaafi nahi — kai indicators use hote hain:

- **Economic indicators:** GDP/GNP, per capita income, savings & investment rate, sectoral composition.
- **Social indicators:** literacy rate, life expectancy, infant mortality rate (IMR), health & nutrition.
- **Composite indices:** HDI (UNDP), PQLI (Physical Quality of Life Index), MPI (Multidimensional Poverty Index), GDI/GII (gender).

1.4 Inclusive Growth

Inclusive Growth wo growth hai jiska fayda society ke **sabhi sections** — especially poor, women, rural aur marginalised — tak pahunche, aur jo **employment & opportunities** broad-based banaye. Sirf high GDP growth kaafi nahi agar wo sirf top par concentrate ho.

India context: 11th & 12th Five Year Plans ka central theme inclusive growth tha; iske liye MGNREGA, financial inclusion (Jan Dhan), skill development, PDS jaise programmes chalaye gaye.

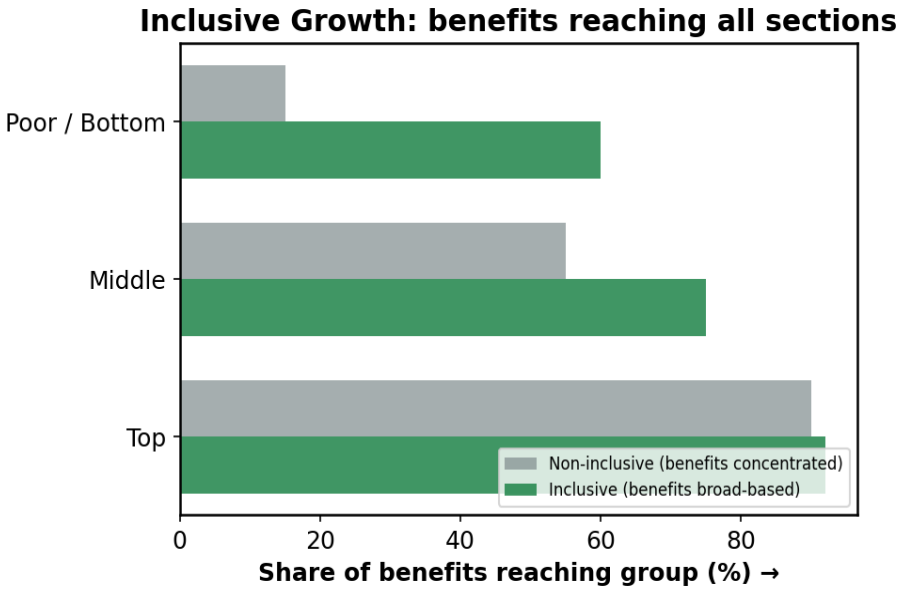


Fig 1.2 — Inclusive growth mein growth ka benefit bottom/poor sections tak bhi pahunchta hai.

1.5 Sustainable Development

Sustainable Development ki famous definition **Brundtland Commission (1987)** se aati hai: *"Development that meets the needs of the present without compromising the ability of future generations to meet their own needs."* Yani aaj ki zaruratein puri karte hue future generations ke resources ko bachaye rakhna.

Iske **teen pillars** hote hain — **Economic, Social aur Environmental** — teeno mein balance zaruri hai. **SDGs (2015):** UN ne 2030 tak ke liye **17 Sustainable Development Goals** diye (poverty, hunger, health, education, clean energy, climate action, etc.).



Fig 1.3 — Sustainable development = Economy + Society + Environment ka overlap (teeno pillars).

2. Circular Causation, Underdevelopment & Measurement

2.1 Circular Causation (Myrdal)

Gunnar Myrdal ka **Circular & Cumulative Causation** principle kehta hai ki social/economic factors ek doosre ko **reinforce** karte hain ek hi direction mein. Underdeveloped economies "**vicious circle of poverty**" mein phans jaati hain: low income → low saving → low investment → low capital → low productivity → phir low income.

Backwash vs Spread effects: Myrdal ke according market forces **backwash effects** (rich regions garib regions ke resources/labour kheech lete hain → inequality badhti) **spread effects** (development padosi regions tak failta) se zyada hote hain — isliye government intervention zaruri hai.

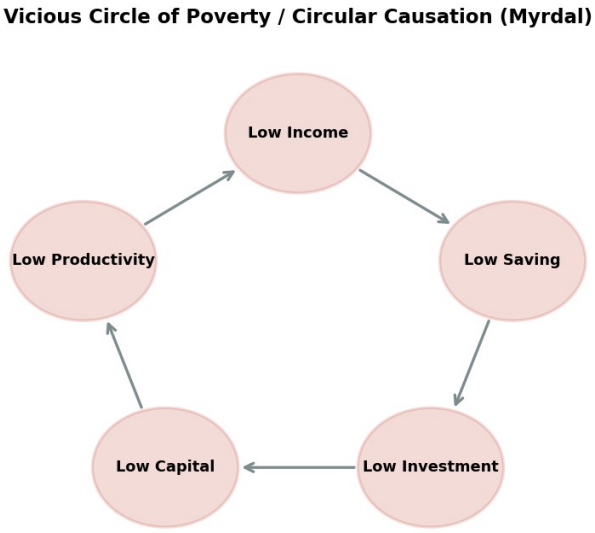


Fig 2.1 — Vicious circle of poverty: har factor agle ko reinforce karta hai (Myrdal).

Vicious Circle — Nurkse

Ragnar Nurkse ne kaha "*a country is poor because it is poor*". Do circles: (i) **Supply side** — low income → low saving → low capital → low productivity → low income; (ii) **Demand side** — low income → low purchasing power → small market → low investment → low income. Inhe todne ke liye balanced 'big' investment chahiye.

2.2 Structural View of Underdevelopment

Structuralist view (Prebisch, Singer, Furtado, Myrdal) kehta hai ki underdevelopment sirf "kam capital" ka masla nahi, balki economy ki **structure** mein hi rukawatein (rigidities) hain:

- **Dualism:** ek modern (industrial/urban) sector aur ek traditional (subsistence agriculture/rural) sector saath-saath — gap badhta hai.
- **Dependence on primary exports:** developing desh kachcha maal export karte hain, manufactured goods import — terms of trade unfavourable.
- **Rigid markets, weak institutions, surplus labour, low mobility** — resources easily shift nahi hote.

Isliye structuralists **state-led planning, industrialisation aur import substitution** ki vakalat karte hain.

2.3 Measurement of Development

(a) Conventional Methods

- **GDP / GNP growth rate** — aggregate output.
- **Per capita income** — income ÷ population (most common, par distribution chhupa deta hai).
- **Sectoral composition** of output & employment (structural change ka sign).

Limitations: inequality, non-market activities, environment aur welfare capture nahi karte (jaise Unit-2 mein discuss kiya).

(b) HDI (Human Development Index)

UNDP (Mahbub-ul-Haq & Amartya Sen) ka **HDI (1990)** teen dimensions ka **geometric mean** hai (value 0 se 1):

1. Long & healthy life — Life Expectancy at birth.
2. Knowledge — Mean years of schooling + Expected years of schooling.
3. Decent standard of living — GNI per capita (PPP).

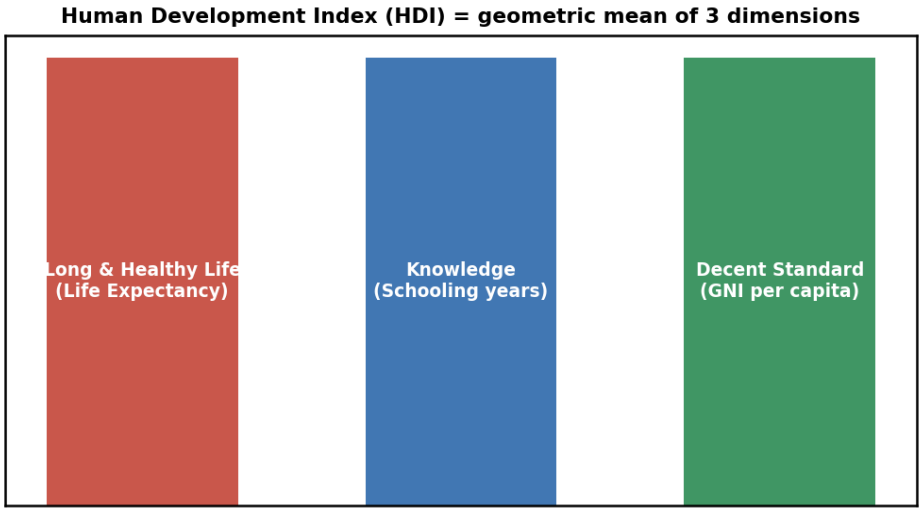


Fig 2.2 — HDI ke teen dimensions: health, knowledge aur standard of living.

(c) Quality of Life Indices

- **PQLI (Morris):** teen indicators ka simple average — life expectancy, infant mortality, literacy (income exclude karta hai).
- **MPI (Multidimensional Poverty Index):** health, education, living standards ke 10 indicators se poverty measure.
- **Other:** Gender Inequality Index (GII), Gross National Happiness (Bhutan), Green GDP, Genuine Progress Indicator.

3. Environment, Climate Change & Coase Solutions

3.1 Environmental Sustainability

Environmental Sustainability ka matlab hai natural resources ka use is tarah karna ki ecosystem ki **regeneration capacity** bani rahe — yani renewable resources ko unke renewal rate se zyada use na karein aur pollution ko absorption capacity ke andar rakhein.

3.2 Environment & Development Dilemma

Developing countries ke saamne ek **dilemma (tension)** hai: tezi se growth/industrialisation chahiye (poverty hatane ke liye), par wahi growth **pollution aur resource depletion** badhati hai. **Environmental Kuznets Curve (EKC)** hypothesis kehti hai ki shuruaat mein income badhne par environmental degradation **badhti** hai, par ek turning point ke baad (jab desh ameer ho jaata hai aur clean tech afford karta hai) degradation **girne** lagti hai — yani inverted-U shape.

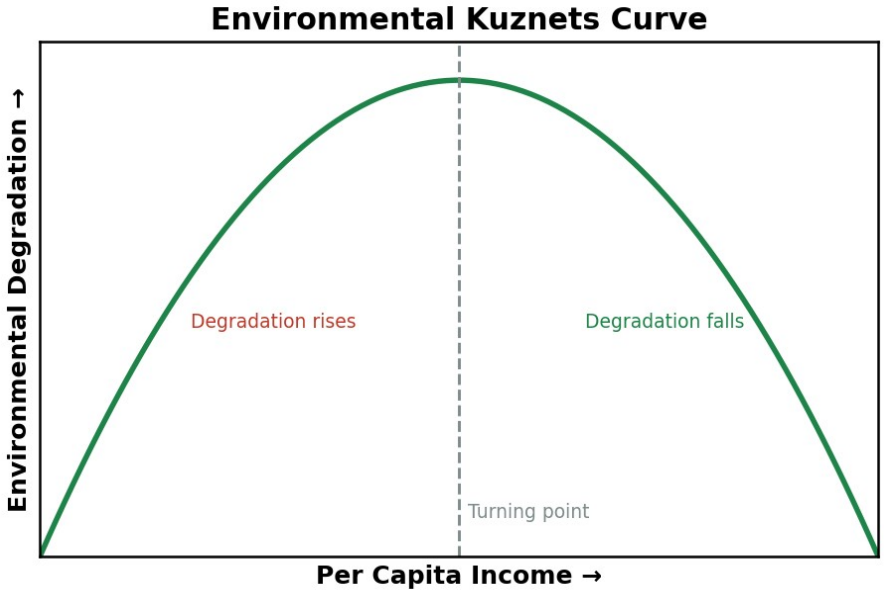


Fig 3.1 — Environmental Kuznets Curve: income badhne par pehle degradation badhti, turning point ke baad ghatti.

3.3 Environmental & Economic Accounting

Normal national income (GDP) environmental damage ko ignore karta hai. **Green Accounting / Environmental Accounting** ismein natural resource depletion aur pollution ki cost ko adjust karta hai — **Green GDP = GDP − cost of environmental degradation & resource depletion.** UN ka framework **SEEA (System of Environmental-Economic Accounting)** hai. India mein "Green GDP"/natural capital accounting ki direction mein kaam ho raha hai.

3.4 Environment Regulation in India

- **Constitutional:** Article 48A (state ko environment protect karna), Article 51A(g) (citizen duty).
- **Key Acts:** Water Act 1974, Air Act 1981, Environment (Protection) Act 1986, Forest Conservation Act 1980, Wildlife Protection Act 1972.
- **Bodies:** CPCB/SPCB (Pollution Control Boards), MoEFCC (Ministry), National Green Tribunal (NGT, 2010) — speedy environmental justice.

- **Instruments:** EIA (Environmental Impact Assessment), emission standards, pollution taxes/cess, carbon markets.

3.5 Climate Change – Impact & Remedies

Climate Change mainly greenhouse gases (CO_2, methane) ke badhne se global warming ki wajah se ho raha hai. **Impacts:** rising temperatures & sea levels, erratic monsoon, droughts/floods, agriculture & food security par asar, biodiversity loss, health risks aur climate migration.

Remedies / Responses:

- **Mitigation:** emissions kam karna — renewable energy (solar/wind), energy efficiency, afforestation, electric vehicles.
- **Adaptation:** change ke saath adjust karna — climate-resilient crops, flood defences, water management.
- **Global agreements:** UNFCCC, Kyoto Protocol (1997), Paris Agreement (2015 — limit warming below 2°C, ideally 1.5°C).
- **India's steps:** National Action Plan on Climate Change (8 missions), International Solar Alliance, Net-Zero by 2070 pledge, LiFE (lifestyle for environment).

3.6 Coase Theorem / Bargaining Solutions

Ronald Coase ne externalities (jaise pollution) ka ek alag solution diya. **Coase Theorem:** agar (i) property rights clearly defined hon aur (ii) transaction costs zero/low hon, to affected parties **aapas mein bargain (negotiate)** karke **efficient outcome** tak pahunch sakti hain — chahe property right kisi ke bhi paas ho — bina government regulation ke.

Example: Ek factory (polluter) aur ek fisherman (victim). Agar bargaining possible ho to dono ek mutually beneficial level of pollution par settle ho jayenge (factory victim ko compensate kar sakti hai, ya victim factory ko pollution kam karne ke liye pay kar sakta hai).

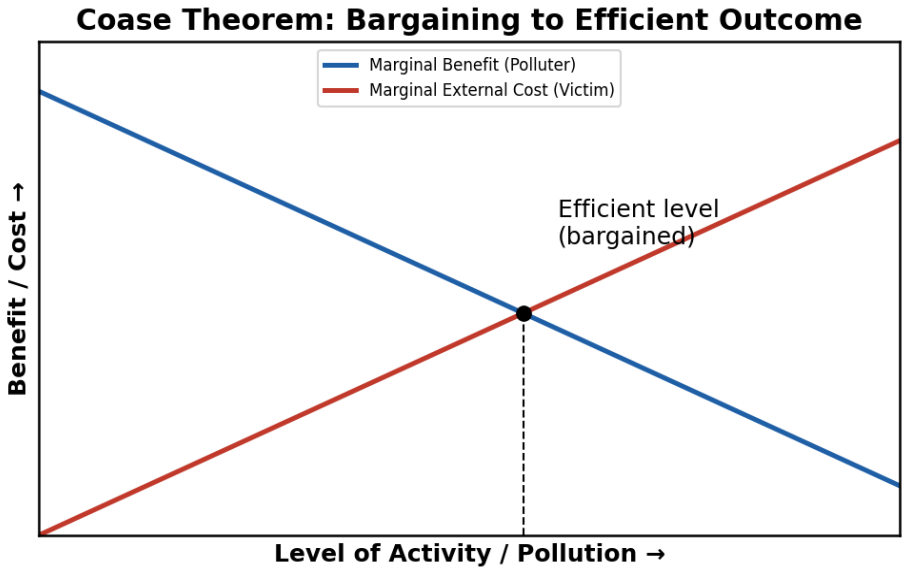


Fig 3.2 — Coase: marginal benefit = marginal external cost par bargaining se efficient activity level.

Coase — Limitations

(i) Real world mein **transaction costs** zyada hote hain. (ii) **Property rights** aksar unclear hote hain (air, oceans). (iii) Bahut saari parties hone par bargaining mushkil (free-rider problem). (iv) Income distribution par depend karta hai. Isliye Pigouvian tax (Unit-1) jaise government solutions bhi zaruri rehte hain.

4. Growth Models

4.1 Harrod–Domar Model

R.F. Harrod aur E. Domar ne (independently) Keynesian ideas ko long-run growth par apply kiya. Core idea: **growth investment (capital accumulation) se aata hai**, aur saving rate + capital-output ratio growth rate decide karte hain.

Growth rate g = s / v
(s = saving rate, v = capital-output ratio / COR)

Yani jitni zyada saving (s) aur jitni kam capital-output ratio (v, yani capital zyada efficient), utni zyada growth. **Implication:** development ke liye saving & investment rate badhana central hai.

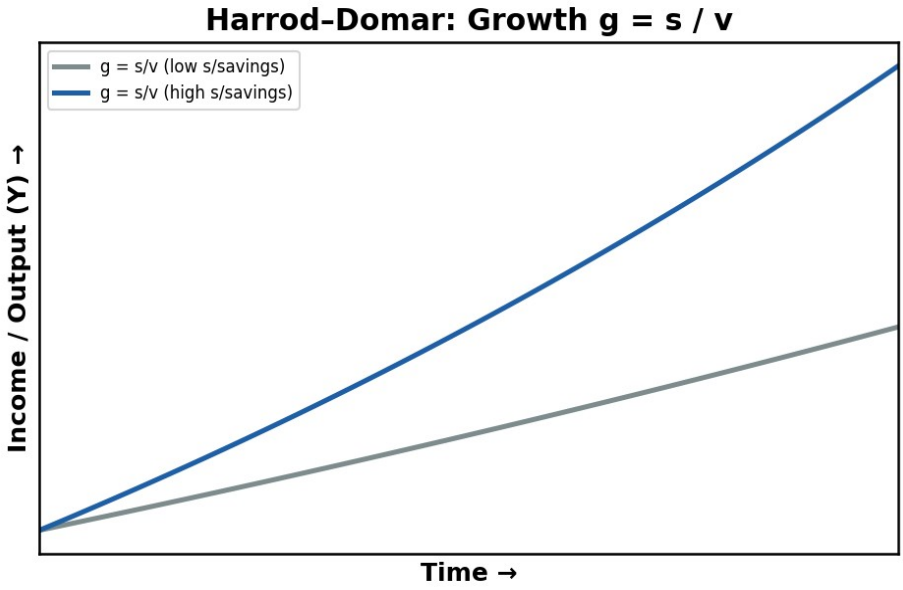


Fig 4.1 — Harrod–Domar: higher saving rate → higher growth path (g = s/v).

Harrod's Three Growth Rates & Knife-Edge

- **Actual growth rate (G):** jo actually realise hoti hai.
- **Warranted growth rate (Gw):** jo entrepreneurs ki expectations ko satisfy karte hue equilibrium maintain kare.
- **Natural growth rate (Gn):** max possible — labour force growth + technical progress se determined.

Knife-edge problem: agar G ≠ Gw, to economy equilibrium se **door bhagti** hai (cumulative boom ya collapse) — growth path bahut unstable ("knife's edge") hai. Long-run full-employment growth ke liye Gw = Gn zaruri, jo apne aap mushkil hai.

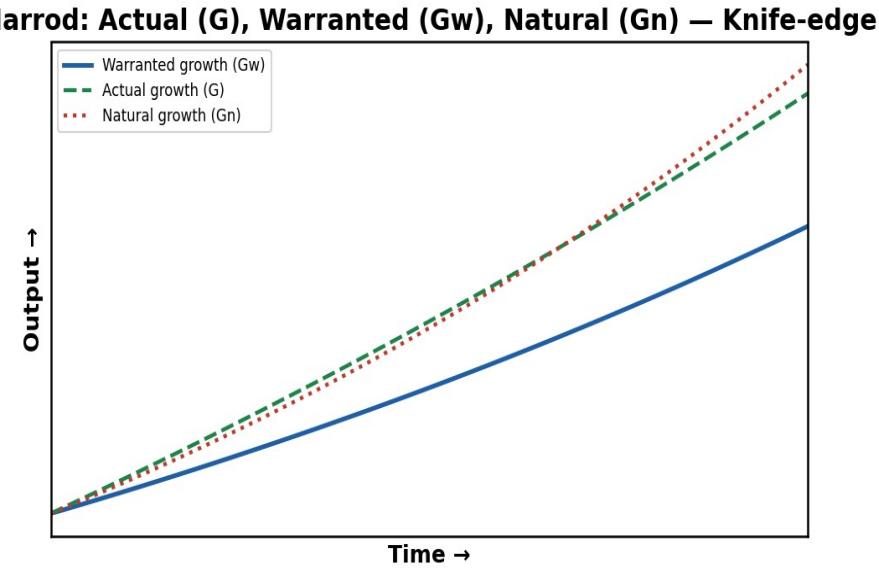


Fig 4.2 — Harrod ki teen growth rates; G aur Gw mein gap unstable 'knife-edge' banata hai.

4.2 Joan Robinson's Model

Joan Robinson (post-Keynesian) ne growth ko **capital accumulation aur income distribution** se jod kar dekha. Unka model dikhata hai ki **rate of accumulation (investment)** aur **profit rate** ek doosre par depend karte hain — zyada accumulation → zyada profit share. Wo ek "**Golden Age**" describe karti hain — wo ideal steady-growth state jahan accumulation smoothly chalti rahe full employment ke saath (jo practice mein mushkil hota hai).

4.3 Kaldor's Model

Nicholas Kaldor ne growth ko **saving aur income distribution** se joda. Key idea: **capitalists (profit earners) ki saving propensity workers se zyada hoti hai (sp > sw).** Isliye jab profit share badhta hai, total saving badhti hai, jo zyada investment & growth support karti hai. Growth aur distribution apas mein jude hain. Kaldor ne "**technical progress function**" bhi di — investment per worker badhne se output per worker badhta hai.

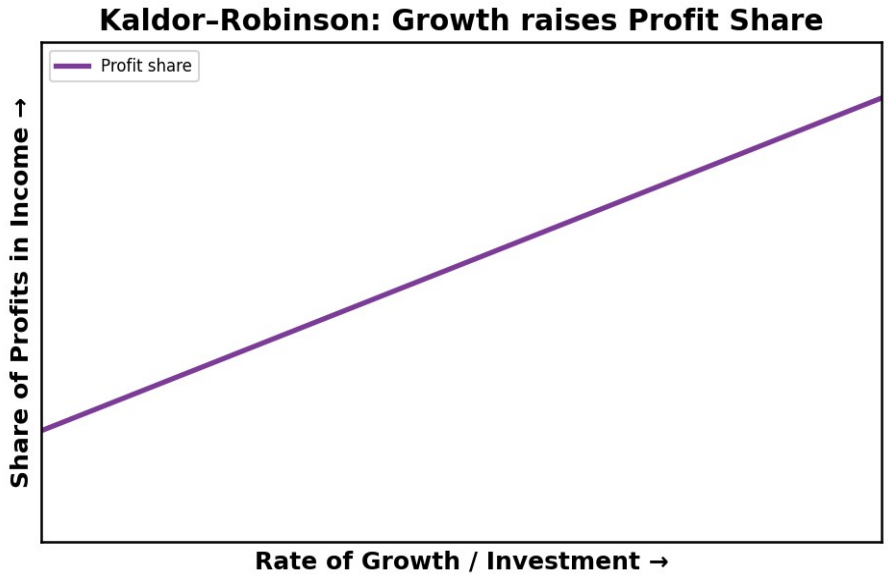


Fig 4.3 — Kaldor/Robinson: higher growth/accumulation → higher profit share → more saving.

4.4 Hicks's Trade Cycle & Technological Progress

J.R. Hicks ne **multiplier–accelerator** interaction (Unit-2) ko growth ke saath joda — output upper limit (full employment 'ceiling') aur lower limit ('floor') ke beech oscillate karta hai, par overall ek growing trend par. Hicks ne technical progress ko classify kiya — **labour-saving, capital-saving aur neutral** — depending on whether it raises MP of capital or labour more. Neutral progress capital-labour ratio constant rakhta hai.

4.5 Meade's Neo-Classical Model

J.E. Meade ne ek **neo-classical growth model** diya jismein output 4 cheezon par depend karta hai — labour force growth, capital accumulation, land aur technical progress. Production function **smooth substitution** allow karta hai (factors ek doosre ki jagah le sakte hain) aur **constant returns to scale** maanta hai. Meade ne dikhaya ki steady growth possible hai aur growth rate factor growth + technical progress par depend karti hai (Solow se milta-julta). Ye Harrod ki 'knife-edge' unstability ko factor substitution se solve karta hai.

4.6 Lewis Model (Unlimited Supply of Labour)

W. Arthur Lewis ka **dual-sector model** do sectors maanta hai — (1) **subsistence/traditional (agriculture)** sector jahan **surplus labour** hai (MP ≈ 0, disguised unemployment), aur (2) **modern/capitalist (industrial)** sector. Agriculture se labour modern sector mein **almost unlimited supply** par (subsistence wage + thoda premium) transfer hota rehta hai.

Modern sector profit kamata hai (capitalist surplus), use **reinvest** karta hai → aur labour absorb hota hai → capital accumulation aur growth. Ye process tab tak chalta hai jab tak surplus labour khatam na ho jaye ("**Lewis Turning Point**"), jiske baad wages badhne lagte hain.

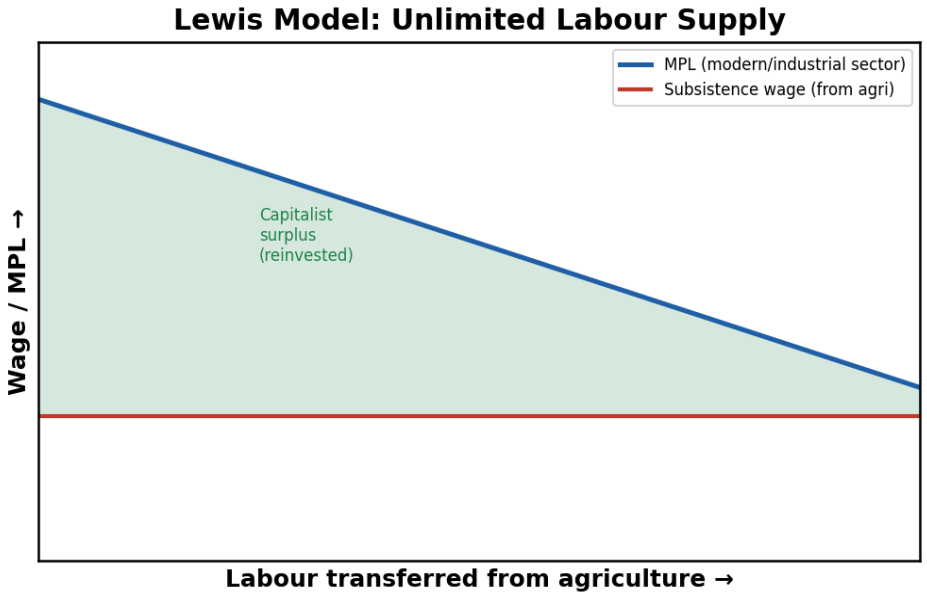


Fig 4.4 — Lewis model: subsistence wage par unlimited labour; modern sector ka surplus reinvest hota hai.

4.7 Ranis–Fei Model

Gustav Ranis aur John Fei ne Lewis model ko extend/refine kiya, agriculture ko zyada importance dete hue. Unke according development **teen phases** mein hota hai:

1. Phase 1: Disguised unemployed labour (MP=0) agriculture se modern sector mein transfer — bina agri output gire.
2. Phase 2: Labour jiska MP > 0 par subsistence wage se kam — transfer jaari, par agri output thoda girta hai (terms of trade shift).
3. Phase 3: 'Commercialisation point' — agri labour ka MP = wage; ab dono sectors fully commercial, surplus labour khatam.

Key contribution: agricultural growth aur productivity ko industrial growth ke liye zaruri bataya — dono sectors ka balanced growth chahiye, warna agri food shortage industrial growth rok degi.

> **Growth Models — One-line Recall**
>
> **Harrod–Domar:** g = s/v; knife-edge instability. **Robinson:** accumulation + distribution; Golden Age. **Kaldor:** distribution-saving link (sp>sw). **Hicks:** multiplier-accelerator cycle, ceiling/floor. **Meade:** neo-classical, factor substitution. **Lewis:** unlimited labour, dual sector. **Ranis–Fei:** Lewis + agriculture, 3 phases.

5. Dependency Theory of Development

Dependency Theory (1950s–60s, Latin America) — main thinkers **Raul Prebisch, Andre Gunder Frank, Paul Baran, Samir Amin.** Ye theory kehti hai ki developing (poor) desh isliye underdeveloped hain kyunki wo **rich (developed) desho par dependent** hain — aur ye dependency historical (colonialism) aur structural hai.

Core–Periphery framework: World do parts mein bata hai — **Core** (developed, industrialised) aur **Periphery** (developing, raw-material producing). Periphery se surplus/resources Core ki taraf flow hota hai → Core aur ameer, Periphery aur garib. Yani "development of the core = underdevelopment of the periphery".

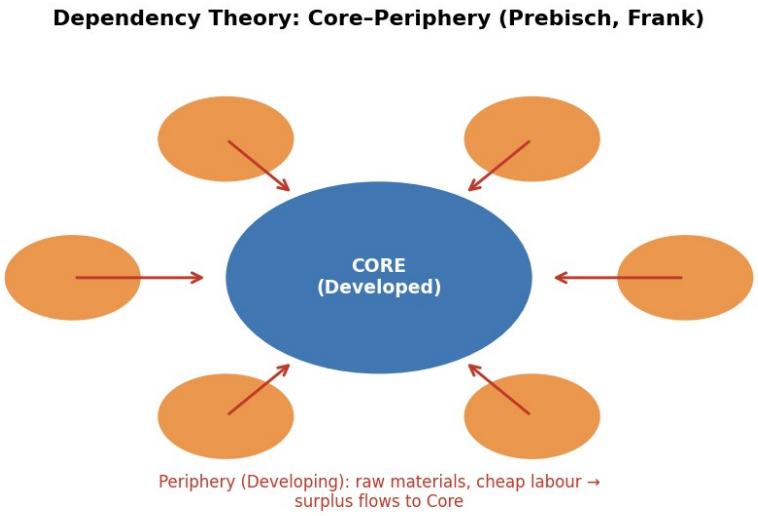


Fig 5.1 — Dependency: periphery se surplus core ki taraf flow hota hai (unequal exchange).

Prebisch–Singer Hypothesis: primary goods (jo periphery export karta hai) ke terms of trade long run mein manufactured goods (core) ke against **girte** hain — isliye trade developing desho ke khilaf kaam karta hai. **Policy suggestion:** import substitution industrialisation (ISI), self-reliance, aur global economic order mein reform. **Criticism:** East Asian 'tigers' ne export-led growth se success paya, jo dependency theory ke against jaata hai.

6. Approaches to Development

6.1 Balanced Growth (Nurkse, Rosenstein-Rodan)

Balanced Growth theory (Ragnar Nurkse, Rosenstein-Rodan) kehti hai ki development ke liye **ek saath, simultaneously kai industries/sectors** mein invest karna chahiye. Reason: ek industry ka product doosri ka market banata hai ("**Say's-law jaisa mutual demand**"). Agar sirf ek industry badhe to uska maal kaun khareedega? Isliye saare sectors ko ek saath upar uthana padta hai taaki market ki "smallness" ki problem hat jaye.

6.2 Unbalanced Growth (Hirschman)

Albert Hirschman ne ulta tark diya — developing desho ke paas resources kam hain, isliye **sab kuch ek saath develop** karna impossible hai. Better hai **kuch 'leading' sectors mein strategically invest** karein, jo apne **linkages** se baaki economy ko aage kheech lein:

- **Backward linkages:** ek industry apne input-supplying industries ki demand badhati hai.
- **Forward linkages:** uska output aage doosri industries ke liye input banta hai.

Deliberately create kiya gaya "imbalance/tension" investment aur growth ko induce karta hai. Public investment **Social Overhead Capital (SOC)** (infra) aur **Directly Productive Activities (DPA)** ke beech alternate karna chahiye.

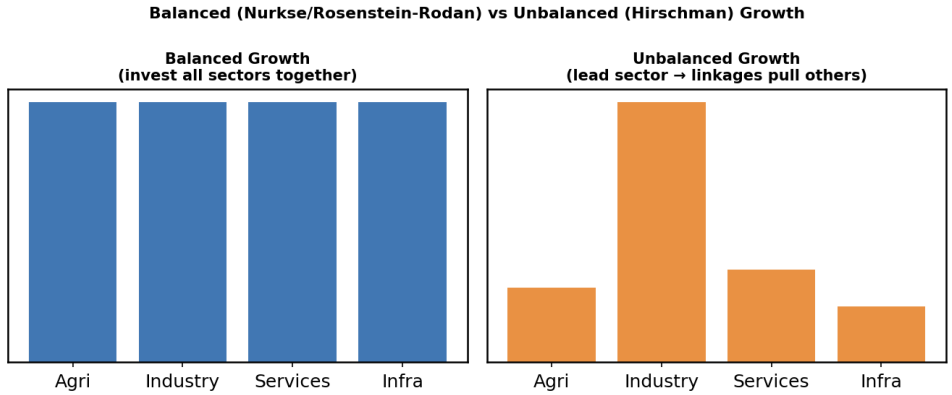


Fig 6.1 — Balanced (sab sectors saath) vs Unbalanced (lead sector linkages se baaki ko kheechta hai).

6.3 Critical Minimum Effort Thesis (Leibenstein)

Harvey Leibenstein ka idea: underdeveloped economy mein **income-raising forces** (investment, tech) aur **income-depressing forces** (mainly population growth) dono kaam karti hain. Choti investment ka asar population growth nigal jaata hai (per capita income wahi rehti). Isliye ek **critical minimum size** ka effort/investment chahiye jo depressing forces se zyada ho aur economy ko **self-sustained growth** par le jaye.

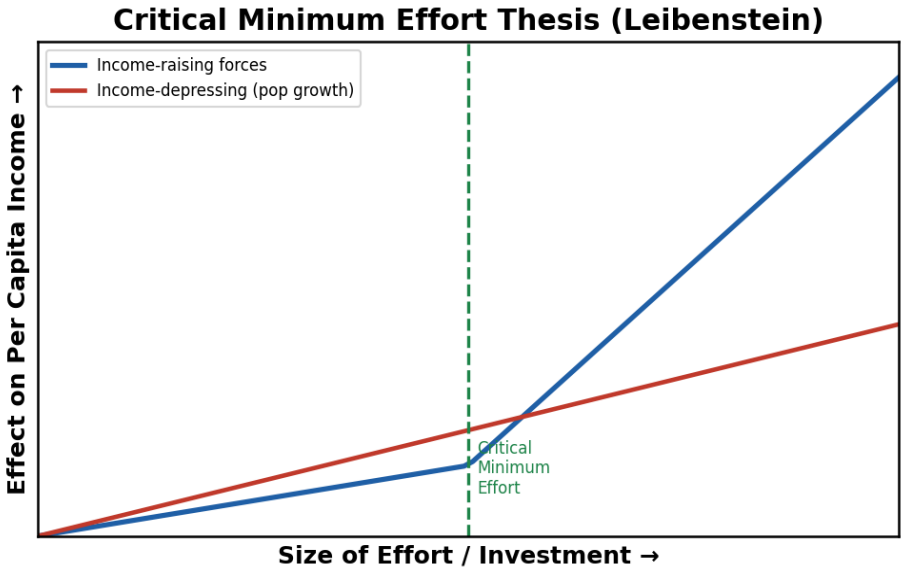


Fig 6.2 — Critical Minimum Effort: ek threshold ke baad income-raising forces population growth ko overtake karti hain.

6.4 Big-Push Theory (Rosenstein-Rodan)

Rosenstein-Rodan ka **Big-Push** theory kehti hai ki development ke liye ek **bada, coordinated, lump-sum (massive) investment** zaruri hai — chhoti-chhoti investments se kaam nahi banega. Iski wajah **indivisibilities (economies of scale)** hain — (i) production function mein (capital lumps mein aata hai), (ii) demand mein (industries ek doosre ke liye market banati hain), (iii) savings mein. In sabko todne ke liye ek hi baar mein 'big push' chahiye taaki external economies generate ho.

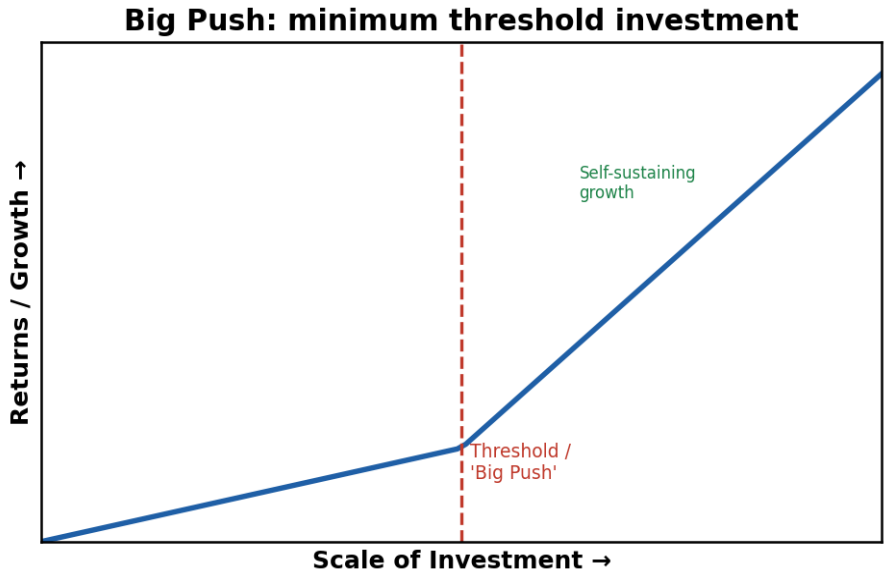


Fig 6.3 — Big Push: threshold scale ke baad hi returns badhte hain aur growth self-sustaining hoti hai.

6.5 Low-Income Equilibrium Trap (Nelson)

R.R. Nelson ka **Low-Level Equilibrium Trap** model dikhata hai ki jab per capita income thodi badhti hai, to **population growth** bhi badh jaata hai (better food/health se death rate girti), jo badhi hui income ko "kha" jaata hai aur per capita income wapas subsistence/low level par aa jaati hai — economy ek **low-income**

equilibrium mein ‘trap’ ho jaati hai. Isse nikalne ke liye income growth ko population growth se kaafi tezi se badhana hoga (critical minimum effort/big push se link).

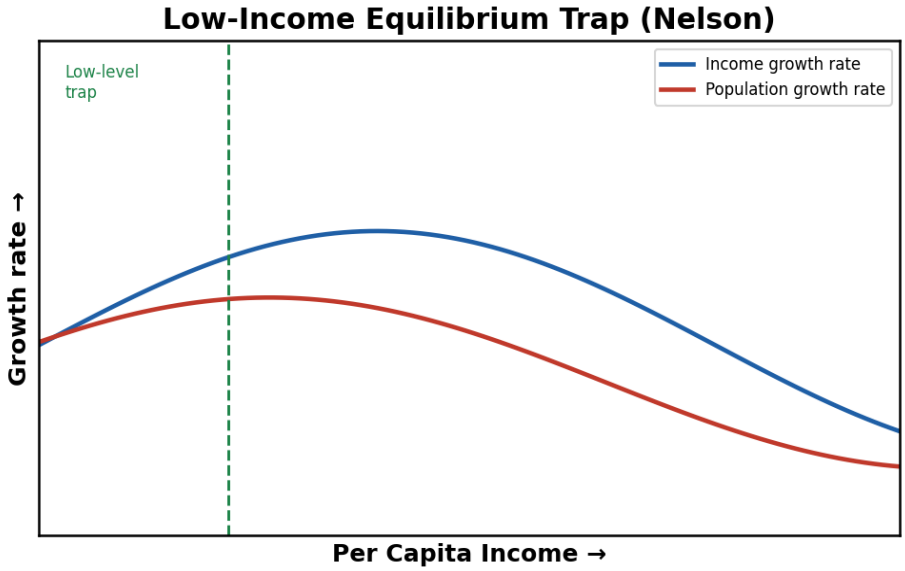


Fig 6.4 — Low-income trap: income badhte hi population growth bhi badh kar use neutralise kar deti hai.

7. Endogenous Growth Theory

7.1 Concept

Neo-classical (Solow) model mein long-run growth **exogenous** technical progress par depend karti thi (model ke bahar se di gayi) — aur capital mein diminishing returns ke kaaran growth ruk jaati. **Endogenous Growth Theory** (1980s, **Paul Romer, Robert Lucas**) kehti hai ki growth economy ke **andar ke factors** (knowledge, R&D, human capital, innovation) se hi aati hai — yani technical progress **endogenous** (model ke andar determined) hai. Knowledge/human capital mein **constant ya increasing returns** hote hain, isliye growth indefinitely sustain ho sakti hai.

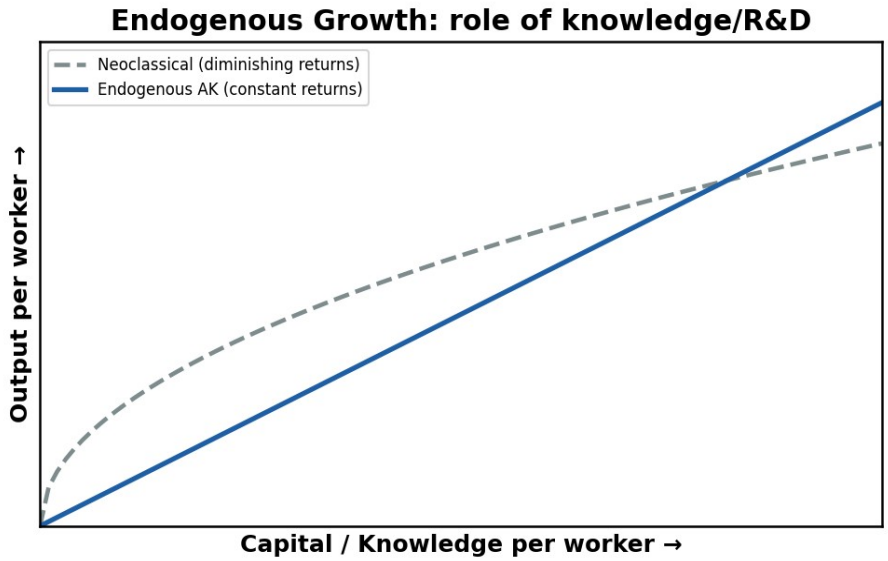


Fig 7.1 — Endogenous (AK) model: capital/knowledge mein diminishing returns nahi → sustained growth.

7.2 Role of Education, Health, Research & Knowledge

- **Education / Human Capital:** skilled, educated workforce productivity badhati hai aur naye ideas generate karti hai (Lucas ne human capital ko central bataya).
- **Health:** healthy workers zyada productive aur lambe samay tak kaam karte hain; health ko bhi human capital ka part maana jaata hai.
- **Research & Development (R&D):** innovation naye products/processes deta hai. Romer ne ideas ko "non-rival" bataya — ek idea bahut log ek saath use kar sakte hain → spillovers.
- **Knowledge spillovers:** ek firm ka knowledge doosri firms ko free mein faayda deta hai (positive externality) — isliye private investment socially optimal se kam hota hai → government ko education/R&D subsidise karna chahiye.

7.3 Explanation of Cross-Country Differentials

Endogenous theory cross-country income/growth differences ko behtar explain karti hai. Solow model predict karta tha ki garib desh ameer desho ko "**catch up (convergence)**" karenge — par real data mein aksar **divergence** dikhta hai (ameer aur ameer, garib aur garib).

Endogenous reasons for differentials: desho ke beech **human capital, R&D investment, technology adoption, institutions, policies aur knowledge access** mein farak hone se growth rates alag rehti hain. Jo desh education, health aur innovation mein zyada invest karte hain, wo sustained higher growth paate

hain — isliye automatic convergence nahi hota ("conditional convergence" — sirf similar fundamentals wale desh converge karte hain).

Endogenous Growth — Policy Takeaways

(i) Education aur skill development mein invest karo. (ii) R&D aur innovation ko subsidise/protect (patents) karo. (iii) Health infrastructure build karo. (iv) Knowledge diffusion aur technology adoption encourage karo. (v) Strong institutions aur policies se sustained growth possible — growth 'choice/policy' ka mamla hai, sirf luck nahi.

— End of Unit 3 : Growth & Development —